

Advanced Software Engineering

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DI INFORMATICA



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Homeworks

Specification of behaviours: genericity

With external description of "deposit«

descriptions.txt contains ["transfer", ["deposit", "withdraw"]]

data.txt contains ["transfer", [0,1], [0, 2]]

```
class Transaction {
    static Map<String,ArrayList<String> descriptions = readSequenceFromFile(descriptions.txt);
    static Map<String,ArrayList<String> data = readSequenceFromFile(data.txt);
    public static boolean execTrans(String transName, Object args[]) {
        boolean check = true; String toDoName; Object[] toDoArgs;
        Iterator<String> desc = getDescription(transName);
        Iterator<String> data = getData(transName);
        Map<String,Object[]> done = new HashTable <String,Object[]>();
        while (desc.hasNext() & data.hasNext & check) {
            toDoName = desc.next(); toDoArgs = extract(args,data.next());
            check = check & PREPARE(toDoName, toDoArgs);
            done.put(toDoName,toDoArgs);
            if !check ROLLBACK(done);
        }
        if check trans.COMMIT(done);
        return check; }
}

Transaction trans = new Transaction(); trans.execTrans("transfer", {amount, otherAcc, custAcc});
```

Homework I

- Try and complete the implementation of the generic mechanism in the previous page

Homework II

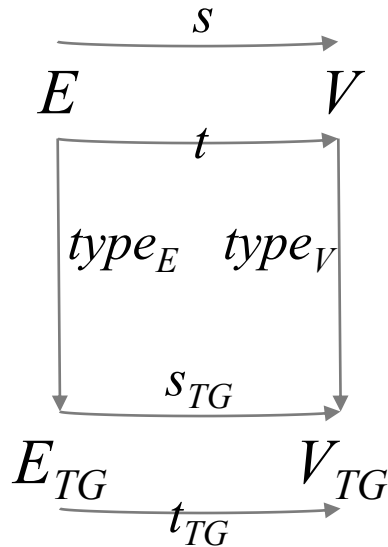
- With reference to slide 34 of Block2Revisiting UML:
- Flatten out the composite state
- Propose a version using fork and joins

Homework III

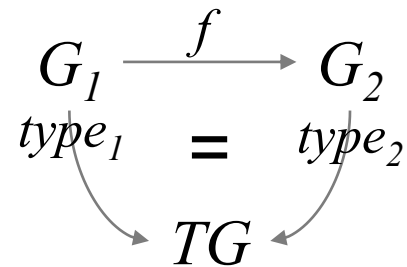
- Exercise on QVT
- Write the transformation from Use Case Diagrams to Class Diagrams using QVT
- Realisable with QVT textual syntax, QVT graphical syntax, or ATL

Homework IV

- Graphs typed over TG
- Prove that it is a category



Typed graph



Typed graph morphism
(f must preserve type)

Labelled graphs can be seen as
special cases of typed graphs

Homework V

- Construct triple graph grammar for transforming Use Case Diagrams to Class Diagrams with AGG.